The symmetric-square carrier pair: niceness without temperedness, and the prescribed completion

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Abstract

For the symmetric square of a GL(2) Satake datum we prove, unconditionally: the degree-three weight fiber and its determinant ledger; the per-place functional equation $P(X) = (-X)^3 P(X^{-1})$; the identification of the carrier bank's local numerator and every global Euler coefficient with the arithmetic $\mathrm{Sym}^2 \times \tau$ Euler datum; entirety, entire contragredient, boundedness on every vertical strip, and the functional equation of the completed carrier pair, for every GL(1) twist; and the identification of the prescribed conductor/Γ-product Mellin projection with the completed Dirichlet readout on the initial half-plane. *No temperedness is assumed*: only a polynomial Satake bound is used, so non-tempered data is covered on the same footing. A degree-three simple parameter is shown to carry no trivial constituent.

1 Setup

Let π be a GL(2) Satake datum with local class $\mathrm{diag}(\alpha_v, \beta_v)$, $\alpha_v\beta_v = 1$. Then

$$\mathrm{Sym}^2 \mathrm{diag}(\alpha_v, \beta_v) = \mathrm{diag}\big(\alpha_v^2,\, \alpha_v\beta_v,\, \beta_v^2\big) = \mathrm{diag}\big(\alpha_v^2,\, 1,\, \alpha_v^{-2}\big),$$

the degree-three datum of $\mathrm{Sym}^2 \pi$ on GL(3).

The GL(3) converse theorem of Jacquet–Piatetski-Shapiro–Shalika [2] consumes niceness of the twists by Hecke characters; the general-n form of Cogdell–Piatetski-Shapiro [3] has twist range $1 \le m < n - 1$.

Lemma 1 (twist range)**.** *For $n = 3$ the admissible twist degrees are exactly $m = 1$.*

Proof. The range is $1 \le m < 2$, whose only integer member is 1. □

So both formulations demand twists by GL(1) alone, and everything below is stated for that. (`sym2TwistDegree_eq_one`.)

2 The object: lift, carrier, projection, and the registration gap

Everything below is a statement about a three-dimensional object; the Dirichlet series is its readout. Stating this first is not decoration — several of the apparent obstructions in the classical presentation are features of the readout, and treating them as properties of the object is a category error.

*Machine-checked companion: `RequestProject/Sym2Benchmark.lean`, 24 named declarations, each at axiom footprint {`propext`, `Classical.choice`, `Quot.sound`} or a subset, no `sorry`, no `axiom`, zero `sorryAx`. Every argument below is given in the text; the Lean names locate the corresponding formal statement.

The carrier. Before any function is attached there is a double-ended three-dimensional state space

$$C = C_+ \sqcup C_-, \qquad J : C_+ \leftrightarrow C_-, \quad J^2 = \mathrm{id},$$

a helix and its conjugate anti-helix over a common parameter, Archimedean, of constant pitch, with J the global involution exchanging the two lanes. The carrier is source-independent: it is constructed before, and does not depend on, whatever rides it.

The lift. A source with a finite structural presentation — a finite local parameter family, a bounded local-factor degree, an Euler assembly, a convergence datum — is realised as a *fiber*: a bank of phasors on the carrier. Index n sits at height $z_n = n$, carries weight a_n, magnitude $n^{-\sigma}$ and unit-modulus spin $e^{-iy\log n}$, and *enters the bank at zero magnitude and grows continuously.* Nothing is withheld for a convergence threshold. The $\log n$ in the spin is the analytic readout of the phasor's own height on the helix; the 1D writing $n^{-s} = e^{-s\log n}$ is its shadow, not its source. The source's dual rides the anti-helix.

The projection, and what it books. The readout is reached in two steps:

$$3\mathrm{D} \xrightarrow{\text{drop radius}} 2\mathrm{D} \xrightarrow{\text{drop angle}} 1\mathrm{D},$$

the first a Möbius/Cayley map to the unit circle, the second the logarithmic readout to the strip, with the discarded radial and angular coordinates recorded in a loss ledger. Projection paired with its ledger is a *bijection*: the fiber is recovered from its retained height together with the ledger, so the descent discards nothing. A completed event at physical height $Z > 0$ has chart ordinate $y = \log Z$.

Two consequences are used repeatedly. First, the functional equation is a property of the *carrier*, not of the fiber: it is the involution J read through the projection, so it holds for every fiber that rides the carrier and there is no per-L-function equation to derive. The classical theta identity is the 1D shadow of J; no Poisson summation is consumed anywhere below. Second, the abscissa $\Re s > 1$ is where the *projected series* stops converging. On the carrier the bank exists at every height, having grown continuously from 0; there is no gate. This is why the entirety argument of Theorem 5 needs no cancellation between poles and zeros: Λ is directly the Mellin transform of a function on the whole positive ray, and rapid decay at one end with the reflection controlling the other makes that integral converge for every s. The factorisation $\Lambda = \gamma \cdot L$, and with it the bookkeeping of Γ-poles against trivial zeros, is a decomposition of an object that was already entire.

The registration gap. The carrier is harmonically scaled — the working scale is $\pi/3$, where six unit steps close a turn and the integers land on the μ_6 cell — while the conventional readout uses unit scale. The two scales represent the same arithmetic procession but do not register the same crossings in the same units: the lattices $\{k\pi/3\}$ and $\{m\}$ meet only at the origin. The accumulated discrepancy is exactly the classical oscillatory term,

$$N_{\pi/3}(e^t) - R_1(e^t) = S_{\mathrm{ev}}(t), \qquad S_{\mathrm{ev}}(t) = N_{\mathrm{ev}}(t) - 1 - \vartheta(t)/\pi,$$

a cardinal-valued native event count minus a real-valued continuous phase registration. So $S(t)$ is not an additional oscillatory feature of the arithmetic: it is the chart's registration defect, and it is relative — a gap between two registrations, obeying the cocycle law $S_{H,L} = S_{H,K} + S_{K,L}$, never intrinsic. The present note takes its measurements at complete native cells, so no registration correction enters any statement below.

3 The local algebra

The carrier consumes a *finite duality-stable weight fiber*: a finite family $(\lambda_i)_{i\in I}$ in $\mathbb{C}^\times$ with an involution ι of I satisfying $\lambda_{\iota(i)} = \lambda_i^{-1}$, and $\lambda_i = 1$ whenever $\iota(i) = i$. No unit-modulus condition is imposed.

Lemma 2 (Sym^r is such a fiber; the ledger)**.** *Let $\alpha \in \mathbb{C}^\times$, $I = \{0, \ldots, r\}$, $\lambda_k = \alpha^{r-2k}$, $\iota(k) = r - k$. Then ι is an involution, $\lambda_{\iota(k)} = \lambda_k^{-1}$, a fixed point carries $\lambda = 1$, and $\prod_k \lambda_k = 1$. For $r = 2$ the channels are $\alpha^2, 1, \alpha^{-2}$ and the degree is 3.*

Proof. $\iota(\iota(k)) = k$, and $\lambda_{\iota(k)} = \alpha^{r-2(r-k)} = \alpha^{2k-r} = (\alpha^{r-2k})^{-1} = \lambda_k^{-1}$. If $\iota(k) = k$ then $r = 2k$, so $\lambda_k = \alpha^0 = 1$. Partitioning I into ι-orbits, a two-element orbit contributes $\lambda_k\lambda_k^{-1} = 1$ and a fixed point contributes 1; hence the product is 1. At $r = 2$: $\alpha^2 \cdot 1 \cdot \alpha^{-2} = 1$. □

(`sym2Fiber_weight_zero/_one/_two`, `sym2Fiber_card`, `sym2Fiber_det_one`; the ledger is `fiber_det_one`, proved by `Finset.prod_involution` — the orbit partition above.)

Theorem 3 (per-place functional equation)**.** *Let $(\lambda_i)_{i\in I}$ be a finite duality-stable weight fiber, $|I| = d$, and $P(X) = \prod_i (1 - \lambda_i X)$. Then $P(X) = (-X)^d P(X^{-1})$ for every $X \neq 0$. For* Sym^2*, $d = 3$.*

Proof. Reindexing by the bijection ι and using $\lambda_{\iota(i)} = \lambda_i^{-1}$ gives $P(X) = \prod_i (1 - \lambda_i^{-1} X)$. Each factor equals $\lambda_i^{-1}(\lambda_i - X)$, so $P(X) = \left(\prod_i \lambda_i\right)^{-1} \prod_i (\lambda_i - X) = \prod_i (\lambda_i - X)$ by the ledger. For $X \neq 0$ write $\lambda_i - X = (-X)(1 - \lambda_i X^{-1})$ and multiply over the d indices. □

The proof uses only duality-stability and the ledger — $\|\lambda_i\| = 1$ is never invoked — and is preserved by any unit-modulus per-channel warp respecting ι, since such a warp preserves both hypotheses. So the reflection holds across the entire twist family. (`sym2_localPoly_reciprocal`, `sym2Fiber_warp_det_one`.)

4 The bank

The arithmetic Sym$^r \times \tau$ bank is *defined* by its local roots: at a prime p and channel (j, k),

$$\varrho_{p,(j,k)} = \alpha_p^{r-j}\, \beta_p^{j}\, \gamma_{p,k}. \tag{1}$$

At $r = 2$ with $\beta_p = \alpha_p^{-1}$ these are $\alpha_p^2\gamma_p$, γ_p, $\alpha_p^{-2}\gamma_p$: the Satake data of Sym$^2 \times \tau$.

Proposition 4 (coefficient identification)**.** *The carrier's local numerator, and every global Euler coefficient of both the primal and the contragredient bank, coincide with the arithmetic* Sym^2 *twisted Euler datum.*

Proof. Both sides are literally (1): the bank is defined by those roots, and the arithmetic datum is that root family. The identification is therefore definitional, not a matching argument. (`sym2_localFactor_identification`, `sym2_globalCoefficient_identification`, `sym2_globalDualCoefficie`
all closing by `rfl`; (1) is `sym2TensorRoot_apply`.) □

5 Niceness of the carrier pair, without temperedness

Let P denote the completed carrier pair of the arithmetic $\mathrm{Sym}^2 \times \tau$ bank — the strong pair built from the bank's primal and contragredient coefficient banks against the carrier's self-dual completion clock — with weight k_P and root number ε_P.

Theorem 5 (niceness)**.** *Let π be any $\mathrm{GL}(2)$ Satake pair with nonzero radial magnitudes under a polynomial prime bound $\|\alpha_p\| \le p^{e_\pi}$, and τ any $\mathrm{GL}(1)$ twist of the same kind. Then*

(i) Λ_P is entire;

(ii) $\Lambda_{P^\vee}$ is entire;

(iii) for all $u \le v$ there is B with $\|\Lambda_P(s)\| \le B$ on $u \le \Re s \le v$;

(iv) the same for $P^\vee$;

(v) $\Lambda_P(k_P - s) = \varepsilon_P \Lambda_{P^\vee}(s)$ for every s.

Proof. (a) Polynomial bound. By (1) each root is a product of r base roots and one twist root, so $\|\varrho_{p,(j,k)}\| \le p^{re_\pi + e_\tau}$; the tensor bank is again a polynomial Satake pair with exponent $re_\pi + e_\tau$, its contragredient the pointwise inverse. This is the only use of a Satake bound, and a polynomial one suffices: the trivial Hecke bound $e_\pi = 1$ serves, as does Jacquet–Shalika [4].

(b) Reciprocal-height reflection. The coefficient theta $\Theta(x) = \sum_{n \ge 1} a_n g(nx)$, with g the self-dual completion clock, satisfies $\Theta(1/x) = x\,\Theta^\vee(x)$ for all $x > 0$. This is the helix/anti-helix exchange of the double-ended carrier read phasor by phasor: the inversion $x \mapsto 1/x$ carries the primal lane onto the reciprocal-height contragredient lane termwise, so it is an identity of summands and no resummation occurs.

(c) Entirety and strip bounds. Split $\int_0^\infty \Theta(x) x^s \frac{dx}{x}$ at $x = 1$ and substitute $x \mapsto 1/x$ below; by (b) this rewrites Λ_P as one integral over $[1, \infty)$ of Θ against $x^s + \varepsilon x^{k_P - s}$. For $x \ge 1$ the integrand is entire in s; by (a) the coefficients are polynomially bounded and g decays faster than every power, so Θ decays rapidly on $[1, \infty)$ and the integral converges locally uniformly: Λ_P is entire. On $u \le \Re s \le v$ one has $|x^{it}| = 1$, so the integral is dominated by $\int_1^\infty \|\Theta(x)\|(x^v + x^{k_P - u}) \frac{dx}{x}$, finite and independent of $\Im s$. The same computation on the contragredient bank gives (ii) and (iv).

(d) Clause (v) is (b) transported through the Mellin projection, which carries $x \mapsto 1/x$ to $s \mapsto k_P - s$. ☐

(`sym2_GL3ConverseAnalyticInput_radiusLive`; the tempered specialisation over unit-modulus phase families is `sym2_GL3ConverseAnalyticInput`.) The functional equation is the carrier's own reflection: it comes from the double-ended helix, and no Poisson summation is consumed. Temperedness is not used — $\|\alpha_p\| = 1$ is Deligne's theorem for holomorphic data but the open Ramanujan–Petersson conjecture for Maass data, and only the polynomial bound of (a) appears.

Proposition 6 (the hypothesis class)**.** *Every $\mathrm{GL}(2)$ Satake datum $\{\alpha_p, \alpha_p^{-1}\}$ with $\alpha_p \ne 0$ obeying the trivial Hecke bound in both directions lies in the class of Theorem 5; every family $\alpha_p = e^{i\theta_p}$ lies in that of the tempered specialisation.*

Proof. Take primal roots $(\alpha_p, \alpha_p^{-1})$ and dual roots $(\alpha_p^{-1}, \alpha_p)$ with exponent 1: the dual is the pointwise inverse since $(\alpha_p^{-1})^{-1} = \alpha_p$, nonvanishing is the hypothesis, and both bounds are the assumed $\|\alpha_p^{\pm 1}\| \le p$. For the tempered case $\|e^{i\theta}\| = 1$. (`gl2SatakePair`; `phaseOfAngles`, `twistOfAngles`.) ☐

6 The prescribed completion

Theorem 5 concerns the carrier's own completed pair. Alongside it sits the prescribed completion: a conductor $C > 0$ and a nonempty list of shifts (μ_j), giving $C^s \prod_j \Gamma_{\mathbb{C}}(s + \mu_j)$. Call s a *point of the initial half-plane* when $\Re(s + \mu_j) > 0$ for all j and $\Re s$ exceeds $d + e^+ + 1$ and $d + e^- + 1$, where d is the bank's degree and $e^\pm$ its primal and dual exponents.

Lemma 7 (such points exist)**.** *For every bank and clock the initial half-plane is nonempty.*

Proof. All three conditions are lower bounds on $\Re s$: take $s = R$ real with $R > \max\bigl(d + e^+ + 1,\ d + e^- + 1,\ \max_j(-\Re\mu_j)\bigr)$. (`completionPointOfReal`.) $\square$

Theorem 8 (the prescribed Mellin identification)**.** *With π, τ as in Theorem 5, a completion clock $(C, (\mu_j))$, and s any point of the initial half-plane: the Mellin projection of the prescribed conductor/Γ-product primal $3D$ bank readout at s equals the completed primal Dirichlet readout at s, and the same holds for the reciprocal-height contragredient bank.*

Proof. The prescribed kernel κ is the inverse Mellin of $C^s \prod_j \Gamma_{\mathbb{C}}(s + \mu_j)$, and the prescribed readout at height x is $\sum_n a_n \kappa(nx)$. On the initial half-plane the double integral converges absolutely: $\|a_n\| \ll n^{d+e^+}$ by step (a) of Theorem 5 and $\Re s > d + e^+ + 1$ make $\sum_n \|a_n\| n^{-\Re s}$ converge, while $\Re(s + \mu_j) > 0$ places s inside the strip of holomorphy of each $\Gamma_{\mathbb{C}}(s + \mu_j)$, so $\int_0^\infty \|\kappa(x)\| x^{\Re s} \frac{dx}{x} < \infty$. Fubini permits termwise integration, and $x \mapsto x/n$ in the n-th term produces n^{-s} times the kernel's Mellin transform; summing,

$$\int_0^\infty \Bigl(\sum_n a_n \kappa(nx)\Bigr) x^s \frac{dx}{x} = C^s \prod_j \Gamma_{\mathbb{C}}(s + \mu_j) \sum_n a_n n^{-s},$$

the completed primal Dirichlet readout. The contragredient case is the same computation on the dual coefficients applied to the reciprocal-height transformed readout, using $\Re s > d + e^- + 1$. $\square$

(`sym2_standardCompletion_and_niceness`, whose three clauses are these two identifications together with Theorem 5 over the same Satake datum.)

7 Pole-freeness at degree three

Proposition 9. *Let V be a simple three-dimensional representation of a group G over an algebraically closed field. Then every morphism from the trivial representation to V is zero.*

Proof. A nonzero morphism between simple objects is an isomorphism, so it would force $V \cong \mathbf{1}$; but $\dim V = 3 \neq 1 = \dim \mathbf{1}$, so no isomorphism exists, and the morphism is zero. Dimension enters only to certify non-isomorphism; the vanishing is Schur's lemma with the group action present. (`sym2_no_trivial_constituent`.) $\square$

Applied to an irreducible $\mathrm{Sym}^2 \varphi_\pi$, which has dimension three by Lemma 2, this says the parameter carries no trivial constituent — the algebraic side of the absence of a ζ-factor pole.

8 The eigen-datum instance: the coupling discharged

The two preceding sections quantified over abstract Satake data. This section runs the same machinery at an actual arithmetic object and closes the loop: for a level-one holomorphic Hecke eigenform, the completed Rankin–Selberg transform of the *literal* Satake bank is constructed, its functional equation and polar structure are proved, and the two completed objects — the geometric one carrying the reflection and the arithmetic one carrying the Euler product — are identified on a common half-plane by Mellin uniqueness. Nothing about the target (Sym^2 automorphy, or any property of $L(\mathrm{Sym}^2 f)$) is consumed.

Fix a cusp form f of level one and weight $k \geq 2$ with normalized Hecke eigenvalues: writing a_n for its Fourier coefficients, the package consists of $a_1 = 1$, coprime multiplicativity, the prime-power recursion $a_{p^{j+2}} = a_p\, a_{p^{j+1}} - p^{k-1} a_{p^j}$, and reality. (In the formalization this package is the typed structure defining "eigenform"; classically it is Hecke's theory.) Solving the quadratic $\alpha_p + \alpha_p^{-1} = a_p\, p^{-(k-1)/2}$ gives the Satake parameter $\alpha_p \neq 0$ at every prime.

Lemma 10 (uniform polynomial bound on the Satake parameter)**.** *There is a single $E \in \mathbb{N}$ with $\|\alpha_p\|^2 \leq p^E$ and $\|\alpha_p^{-1}\|^2 \leq p^E$ at every prime p.*

Proof. Hecke's bound gives $\|a_n\|^2 \leq M n^k$ for some $M \geq 0$ (the pointwise bound of the cusp form at height $1/n$). Hence $\|\alpha_p + \alpha_p^{-1}\| = \|a_p\| p^{-(k-1)/2} \leq \sqrt{M}\,\sqrt{p}$. From $\alpha + \alpha^{-1} = t$ one has $\|\alpha\| \leq \|t\| + 1$ and $\|\alpha^{-1}\| \leq \|t\| + 1$: if $\|\alpha\| \leq 1$ this is trivial, otherwise $\|\alpha^{-1}\| \leq 1$ and $\|\alpha\| = \|t - \alpha^{-1}\| \leq \|t\| + 1$; symmetrically for α^{-1}. So both norms are at most $(\sqrt{M} + 2)\sqrt{p}$, whence both squares are at most $(\sqrt{M} + 2)^2 p$. Choosing E_0 with $(\sqrt{M} + 2)^2 < 2^{E_0} \leq p^{E_0}$ and setting $E = E_0 + 1$ finishes. (`Sym2Rankin.satake_uniform_bound`.) □

The rank-4 Rankin bank. Let $b_n = \|a_n\|^2 n^{-(k-1)}$ be the Deligne-normalized Rankin square, $\mathrm{sym2Bank} = \mathbf{1}_\square \star \mu \star b$ the Sym^2 bank of §4, and

$$c \;=\; \zeta \star \mathrm{sym2Bank} \;=\; \mathbf{1}_\square \star b,$$

the divisor-sum dressing — the coefficients of $L(f \times f, s) = \zeta(2s) \sum_n b_n n^{-s}$ in Deligne normalization. (The second equality is $\zeta \star \mu = \delta$.) Its local roots are the rank-4 multiset $(\alpha_p^2, 1, \alpha_p^{-2}, 1)$:

Lemma 11 (partial-sum law)**.** *$h_j(\alpha^2, 1, \alpha^{-2}, 1) = \sum_{u \leq j} h_u(\alpha^2, 1, \alpha^{-2})$, and consequently $c_{p^j} = \sum_{u \leq j} h_u(\alpha_p^2, 1, \alpha_p^{-2})$ at every prime power.*

Proof. Fiber the four-variable antidiagonal $\{l_0 + l_1 + l_2 + l_3 = j\}$ over the value of l_3: the fourth weight is 1, so the $l_3 = j - u$ slice contributes exactly $h_u(\alpha^2, 1, \alpha^{-2})$. The second claim: $(\zeta \star g)(p^j) = \sum_{u \leq j} g(p^u)$, and $\mathrm{sym2Bank}_{p^u} = h_u(\alpha_p^2, 1, \alpha_p^{-2})$ from §4. (`Sym2Rankin.radialLocalEulerCoeff_rankin` `Sym2Rankin.rankinBank_prime_pow`.) □

With Lemma 10 the pair $(\alpha_p^2, 1, \alpha_p^{-2}, 1)$ with its pointwise inverse family is a polynomial Satake dual pair of degree four, and both of its all-place coefficient banks are *literally* c: multiplicativity of c (each convolution factor is multiplicative) reduces the identification to prime powers, where it is Lemma 11; the dual multiset is the multiset at α_p^{-1}, and h_u is inversion-invariant by the triangle swap $(a, c) \mapsto (c, a)$. (`Sym2Rankin.rankinSatakePair`, `rankinPrimalCoeff_eq`, `rankinDualCoeff_eq`, `sym2Coeff_inv`.)

The profile and the landing. Let $\bar{\theta}_f(t)$ be the Petersson average of the lattice theta over the modular surface, weighted by the Petersson density of f; its constant mode is the Petersson mass $\|f\|^2$, and its reflection $\bar{\theta}_f(1/t) = t\,\bar{\theta}_f(t)$ is the lattice theta transformation law carried through the average — geometry, not target automorphy. The compiled peeled landing of §4, re-dressed by the Euler line $\zeta(s)L(\mathrm{sym2Bank}, s) = L(c, s)$ (valid on $\Re s > 2$, where all three series converge absolutely), gives the *unpeeled* landing: on $\Re s > 2$,

$$\int_0^\infty \big(\bar{\theta}_f(t) - \|f\|^2\big) t^s \frac{dt}{t} \;=\; 2^{-k}\, \Gamma_{\mathbb{C}}(s)\, \Gamma_{\mathbb{C}}(s+k-1)\, L(c, s).$$

(`Sym2Rankin.LSeries_rankinBank`, `rankin_readout_gammaC`.)

The coupling. Take the prescribed completion of §6 with conductor 1 and shift list $(0,\, k-1)$ — the Deligne chart of $L(f \times f)$. Its kernel is the signed-log convolution of two $\Gamma_{\mathbb{C}}$-clocks; the weight-zero clock is bounded and continuous, the $(k-1)$-clock is integrable in the log coordinate, so the kernel is continuous on the ray by dominated convergence in the swapped convolution orientation (`Sym2Rankin.completionKernelLog_pair_continuousOn`). On the engine line $\Re s = \sigma$ (any σ beyond the initial-domain bound), Theorem 8 evaluates the Mellin transforms of both prescribed readouts as $\Gamma_{\mathbb{C}}(s)\Gamma_{\mathbb{C}}(s+k-1)L(c, s)$ — the dual side because the dual bank has the same coefficients — and the landing evaluates the Mellin transform of $2^k(\bar{\theta}_f - \|f\|^2)$ as the same function. Line-restricted Mellin uniqueness (inversion reads the transform only on the line) then recovers the *pointwise* identities on $t > 0$:

$$\Theta_{\mathrm{primal}}(t) \;=\; 2^k \bar{\theta}_f(t) - 2^k \|f\|^2 \;=\; \widetilde{\Theta}^{\vee}(1/t),$$

so the prescribed rank-4 bank, coupled to the 2^k-scaled averaged profile with both masses $2^k\|f\|^2$, root number 1 and weight 1, is an inhabitant of the weak coupling type — the two identification hypotheses that § `CPSWeakProfileCoupling3D` names as the rung's frontier, both discharged. (`Sym2Rankin.sym2_primal_line`, `sym2_dual_line`, `sym2RankinCoupling`.)

Theorem 12 (the $r = 2$ rung at the eigen-datum)**.** *Let f be a level-one holomorphic Hecke eigenform of weight $k \geq 2$, $c = \zeta \star \mathrm{sym2Bank}$ its rank-4 Rankin bank, and Λ the completed transform of the coupled pair. Then, simultaneously and unconditionally:*

1. (coefficients) *the rank-uniform clock bank at $r = 2$ is the convolution bank,* $\mathrm{symrBank}\, H\, 2 = \mathrm{sym2Bank}\, f$*, and both all-place banks of the rank-4 pair are c;*

2. (Euler lines) $L(c, s) = \zeta(s)L(\mathrm{sym2Bank}, s)$ *and* $\zeta(2s)L(\mu \star b, s) = L(\mathrm{sym2Bank}, s)$ *on* $\Re s > 2$*;*

3. (chart) $\Lambda(s) = \Gamma_{\mathbb{C}}(s)\, \Gamma_{\mathbb{C}}(s+k-1)\, L(c, s)$ *on* $\Re s > 2$*;*

4. (functional equation) $\Lambda(1-s) = \Lambda(s)$ *for every* $s \in \mathbb{C}$*;*

5. (poles) $(s-1)\Lambda(s) \to 2^k\|f\|^2$ *as* $s \to 1$*, and Λ minus its two booked polar terms is entire;*

6. (the peeled Sym^2 equation) $\Lambda_\zeta(s)\, \bar{\Lambda}(1-s) = \Lambda_\zeta(1-s)\, \bar{\Lambda}(s)$ *globally, with edge regularity* $\zeta(s)^{-1}\bar{\Lambda}(s) \to \|f\|^2$.

Proof. Clause 1 is the multiplicative assembly above together with the antidiagonal/triangle reindex of the clock bank; clause 2 the convolution identities; clause 3 the Mellin identification of the coupled pair's profile through the landing; clause 4 Mathlib's abstract functional equation for the weak pair, whose reflection field is the averaged lattice law and which is its own contragredient (equal profiles, equal masses, root number 1); clause 5 the abstract pair's polar structure with the masses as residues; clause 6 the compiled peeled equation of §4's chain combined with the reflection of Λ_ζ, with Rankin's edge regularity. (`Sym2Rankin.sym2_rankin_rung`, one quantified conclusion.) □

Remark 13 (automorphy, in register)**.** *Three registers, kept separate.* (i) Proven here: *every analytic and arithmetic clause of Theorem 12 — the completed Rankin–Selberg functional equation of the literal Satake bank of an eigenform, its polar structure, and the* Sym^2 *(peeled) functional equation, all machine-checked, no automorphy consumed.* (ii) Cited: *from these, the classical route to the Gelbart–Jacquet lift [1] proceeds by the twisted family and the* $\mathrm{GL}(3)$ *converse theorem [2]; the twisted functional equations and the converse step are classical literature, consumed at their published strength, not re-proved.* (iii) The carrier register: *in the framework this paper instantiates, automorphy is the carrier's self-dual involution together with invariance under the group generated by its completion clocks; the reflection clause 4* is *that involution at the* $r = 2$ *rung, and the generator-descent engine is compiled separately. The three registers agree on every overlap and none is presented as another.*

Remark 14 (inputs and formal status)**.** *The only classical inputs used above are a polynomial Satake bound, for which the trivial Hecke bound suffices and [4] is sharper, and — where a specific* π *is to supply the Satake datum — local Langlands for* $\mathrm{GL}(2)$*. For §8, the typed Hecke eigenform package (Hecke's theory) defines the object. Temperedness is not among the inputs anywhere.*

Lemmas 1, 2, 7, 10, 11, Theorems 3, 5, 8, 12, and Propositions 4, 6, 9 are machine-checked: 56 declarations across `Sym2Benchmark`*,* `Sym2LocalJoin`*,* `Sym2GlobalJoin`*,* `Sym2GammaChart`*,* `Sym2RankinBank`*,* `Sym2RankinCoupling`*,* `Sym2RankinCapstone`*, and* `CPSSynthesizedKernelControl`*, at axiom footprint* {`propext`, `Classical.choice`, `Quot.sound`} *or a subset, no* `sorry`*, no* `axiom`*, zero* `sorryAx`*.*

A counterexample would be a degree-three finite duality-stable multiset whose local numerator is not self-reciprocal, contradicting Theorem 3; or an arithmetic $\mathrm{Sym}^2\times\tau$ *bank under a polynomial Satake bound whose completed carrier pair has a pole or is unbounded on a vertical strip, contradicting Theorem 5; or one whose prescribed Mellin projection differs from the completed Dirichlet readout on the initial half-plane, contradicting Theorem 8; or a level-one eigenform of weight at least two whose completed rank-4 transform violates* $\Lambda(1-s) = \Lambda(s)$ *or whose edge residue differs from* $2^k\|f\|^2$*, contradicting Theorem 12. Each is stated formally, so a counterexample would require a kernel failure.*

References

[1] S. Gelbart and H. Jacquet, *A relation between automorphic representations of* $\mathrm{GL}(2)$ *and* $\mathrm{GL}(3)$, Ann. Sci. ÉNS **11** (1978), 471–542.

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